1.6 Describing Pairs of Angles

Name _____ Period ____ Date _____

Geometry 1.6 · Student sheet

I can…

- identify complementary and supplementary angles and find their measures.
- use linear pairs to find angle measures.
- use vertical angles to find angle measures.

Vocabulary · fill in the missing word

Complementary angles — two angles whose _____ have a sum of 90°

Supplementary angles — two angles whose _____ have a sum of 180°

Adjacent angles — two angles that share a common vertex and side, but have no common _____ points

Linear pair — two adjacent angles whose _____ sides are opposite rays

Vertical angles — two angles whose sides form two pairs of _____ rays

Part 1 · Discovery: two crossing lines

Draw two lines that cross. Number the angles 1, 2, 3, 4 going around. Measure each one.

m∠1	m∠2	m∠3	m∠4	m∠1 + m∠2	m∠2 + m∠3

Which angles are equal? _____ What do angles next to each other add to? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Sums: complementary angles add to _____; supplementary angles add to _____.

Example 1

Find the complement and the supplement of a 62° angle.

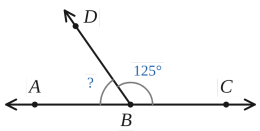
Example 2

∠A and ∠B are supplementary. m∠A = (4x + 8)° and m∠B = (x + 2)°. Find each measure.

Linear Pair Postulate: if two angles form a linear pair, then they are _____.

Example 3

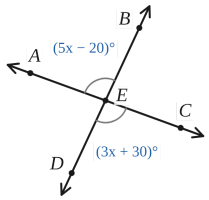
∠1 and ∠2 form a linear pair. m∠1 = 125°. Find m∠2.



Vertical Angles Congruence Theorem: vertical angles are _____.

Example 4

Two lines intersect. Find x, then find all four angle measures.



Example 5

Two angles form a linear pair. One angle is 4 times as large as the other. Find both angles.

Part 3 · Partner practice

Partner 1

Find the complement and the supplement of a 47° angle.

Partner 2

Two vertical angles measure (7x − 9)° and (4x + 21)°. Find x, the angle measure, and the measure of an angle adjacent to it.